

AREEBA ALI

TCT - 004

MT - 227

Find the order, degree, linear, non-linear, ODE, PDE of following equations.

a) $e^x dx + e^y dy = 0$

$$e^x + e^y \frac{dy}{dx} = 0$$

order = 1

degree = 1

linear

ODE

b) $y'' + n^2 y = 0$

order = 2

degree = 1

linear

ODE

c) $y = xy' + x/y'$

$$y = x \frac{dy}{dx} + \frac{x}{\frac{dy}{dx}}$$

order = 1

degree = 2

Non-linear

ODE

d) $[1 + (y')^2]^{3/2} = c^2 (y'')^2$

$$\sqrt{[1 + (y')^2]^3} = (c^2 (y'')^2)^{1/2}$$

$$1 + (y')^2 = c^4 (y'')^4$$

order = 2

degree = 4

Non-linear

ODE

e) $y = xy' + a[1 + (y')^2]$

order = 1

degree = 2

Non-linear

ODE

f) $y'''' + 4y''' + 6y'' + y' = \cos x$

order = 4

degree = 1

linear

ODE

g) $U_{xx} + U_{xy} = 0$

order = 2

degree = 1

linear

PDE

$$h) (y')^2 = (y'' + y)^{3/2}$$

$$((y')^2)^2 = \sqrt{(y'' + y)^3}$$

$$(y')^4 = (y'' + y)^3$$

$$\text{order} = 2$$

$$\text{degree} = 3$$

Non-linear

ODE

Q2

Verify that

a) $y = a + be^{-2x} + e^x/3$ is the general sol of $y'' + 2y' = e^x$

Solution

First find y' and y''

$$y' = \frac{d}{dx} \left(a + be^{-2x} + \frac{1}{3}e^x \right)$$

$$y' = 0 + -2be^{-2x} + \frac{1}{3}e^x$$

$$y' = -2be^{-2x} + \frac{1}{3}e^x$$

$$y'' = \frac{d}{dx} \left(-2be^{-2x} + \frac{1}{3}e^x \right)$$

$$y'' = +4be^{-2x} + \frac{1}{3}e^x$$

Now substitute y' and y'' in $y'' + 2y' = e^x$

$$4be^{-2x} + \frac{1}{3}e^x + 2(-2be^{-2x} + \frac{1}{3}e^x) = e^x$$

$$4be^{-2x} + \frac{1}{3}e^x + -4be^{-2x} + \frac{2}{3}e^x = e^x$$

$$(4b-4b)e^{-2x} + \left(\frac{1}{3} + \frac{2}{3}\right)e^x = e^x$$

$$0 \cdot e^{-2x} + 1 \cdot e^x = e^x$$

$$e^x = e^x$$

$$L.H.S = R.H.S \quad \text{Answer}$$

FORMATION OF D.E

3
a)

$$y = ax^3 + bx^2$$

Solution

differentiate w.r.t x

$$y' = 3ax^2 + 2bx$$

differentiate again

$$y'' = 6ax + 2b$$

$$y'' = 6ax + 2b$$

$$\frac{d^2y}{dx^2} = 6ax + 2b$$

$$\boxed{\frac{y'' - 2b}{6x} = a}$$

substitute 'a' into y' eq

$$y = \frac{1}{3} \left(\frac{y'' - 2b}{6x} \right) x^2 + 2bx$$

$$y = x \left(\frac{y'' - 2b}{2} \right) + 2bx$$

$$y = x \left(\frac{y''}{2} - \frac{2b}{2} \right) + 2bx$$

$$y = \frac{xy''}{2} - \frac{2bx}{2} + 2bx$$

$$y = \frac{xy''}{2} - bx + 2bx$$

$$y = \frac{xy''}{2} + bx$$

$$\boxed{b = \frac{y}{x} - \frac{xy''}{2}}$$

plug into $y = ax^3 + bx^2$

$$y = x^3 \left(\frac{y'' - \frac{y'}{x}}{3x} \right) + x^2 \left(\frac{y'}{x} - \frac{xy''}{2} \right)$$

$$y = \frac{x^2}{3} \left(y'' - \frac{y'}{x} \right) + x \left(y' - \frac{xy''}{2} \right)$$

$$y = \frac{x^2 y''}{3} - \frac{x^2 y'}{3x} + xy' - \frac{x^2 y''}{2}$$

$$y = \frac{x^2 y''}{3} - \frac{xy'}{3} + xy' - \frac{x^2 y''}{2}$$

$$y = \left(\frac{1}{3} - \frac{1}{2} \right) x^2 y'' + \left(-\frac{1}{3} + 1 \right) xy'$$

x b.s by 6

$$6y = -x^2 y'' + 4xy'$$

$$\boxed{x^2 y'' - 4xy' + 6y = 0}$$

Answer

$$xy = Ae^x + Be^{-x}$$

Solution

∴ b.s by x

$$y = \frac{Ae^x + Be^{-x}}{x}$$

differentiate b.s w.r.t x

$$(xy)' \frac{d}{dx} = \left(\frac{Ae^x + Be^{-x}}{x} \right) \frac{d}{dx}$$

$$\frac{dy}{dx} x = (Ae^x + Be^{-x}) \frac{d}{dx}$$

$$y'x + y = Ae^x + Be^{-x}$$

differentiate again

$$xy'' + y' + y' = Ae^x + Be^{-x}$$

$$2y' + xy'' = \underbrace{Ae^x + Be^{-x}}_{xy}$$

$$2y' + xy'' = xy$$

$$xy'' + 2y' - xy = 0 \quad \underline{\text{Answer}}$$

c)

$$y^2 = 4a(x+a)$$

Solution

differentiate

$$\frac{d}{dx} y^2 = \frac{d}{dx} (4a(x+a))$$

$$\frac{d}{dx} y^2 = \frac{d}{dx} (4ax) + \frac{d}{dx} \cdot 4a^2 \quad \frac{da}{dx} = 0$$

$$\frac{d}{dx} y^2 = 4a \cdot \frac{d}{dx} x + 0$$

$$2y y' = 4a$$

$$a = \frac{(2yy')}{2 \cdot 4}$$

$$\boxed{a = \frac{yy'}{2}}$$

put value of a in eq

$$y^2 = 4 \left(\frac{yy'}{2} \right) \left(x + \frac{yy'}{2} \right)$$

$$y^2 = 2yy'x + y^2 y'^2$$

$$y^2 - 2xyy' - y^2 y'^2 = 0 \quad \underline{\text{Answer}}$$

÷ by y

$$y - 2xy' - yy'^2 = 0$$

$$y = ae^{2x} + be^{-3x} + ce^x$$

Solution:

differentiate w.r.t x

$$\frac{dy}{dx} = 2ae^{2x} - 3be^{-3x} + ce^x$$

differentiate again

$$y'' = 4ae^{2x} + 9be^{-3x} + ce^x$$

diff again

$$y''' = 8ae^{2x} - 27be^{-3x} + ce^x$$

using operator method to eliminate constants

$$\text{roots} = 2, -3, 1$$

$$(D-2)(D+3)(D-1)y = 0$$

$$\Rightarrow \text{---} \text{---} \text{---}$$

$$(D-2)(D+3) = D^2 + 3D - 2D - 6 \rightarrow (D^2 + D - 6)(D-1)$$

$$\rightarrow D^3 - D^2 + D^2 - D - 6D + 6$$

$$D^3 - 7D + 6$$

$$\boxed{y''' - 7y' + 6y = 0} \quad \text{Answer}$$

e) $Ax^2 + By^2 = 1 \rightarrow \text{eq 1}$

Solution

Differentiate

$$2ax + 2Byy' = 0$$

$$ax + Byy' = 0 \rightarrow \text{eq 2}$$

$$ax = -byy' \rightarrow \text{eq 3}$$

$$\boxed{a = -\frac{byy'}{x}} \rightarrow \text{eq 3}$$

substitute a into eq 1

$$\left(-\frac{byy'}{x}\right)x^2 + By^2 = 1$$

$$-byy'x + by^2 = 1$$

$$b(-xyy' + y^2) = 1$$

$$\boxed{b = \frac{1}{-xyy' + y^2}}$$

put b into eq 3

$$a = -\frac{byy'}{x} = -\frac{yy'}{x(y^2 - yy'x)}$$

~~diff eq 1 again~~

$$\text{denominator } D(x) = y^2 - xyy'$$

$$\frac{d}{dx}(D(x)) = 0, D'(x) = 0 \text{ provided } D(x)$$

can't substitute A & B back in original eq 1 beca A & B are terms can't depend on x , so expression must be independent of x

$$D'(x^2) = 2xy'$$

$$\frac{d}{dx}(xyy') = 1 \cdot yy' + x \cdot \frac{d}{dx}(yy')$$

$$\frac{d}{dx}(xyy') = yy' + x(y'^2 + yy'')$$

$$D'(x) = 2yy' - (yy' + x(y'^2 + yy''))$$

$$\boxed{yy' - xy'^2 - xyy'' = 0} \quad \text{Answer}$$

$$y = a \sin x + b \cos x + x \sin x \quad \rightarrow \text{eq 1}$$

solution

$$y' = +a \cos x - b \sin x + (\sin x + x \cos x)$$

$$y'' = -a \sin x - b \cos x + (\cos x + -x \sin x + \cos x)$$

$$y'' = -a \sin x - b \cos x + (2 \cos x - x \sin x) \quad \rightarrow \text{eq 3}$$

eliminating from eq 1

$$y - x \sin x = a \sin x + b \cos x \quad \rightarrow \text{eq 2}$$

diff w.r.t x

$$y' - (\sin x + x \cos x) = a \cos x - (\sin x)$$

To eliminate ~~eq 3~~ ^{eq 2} ~~eq 3~~ ^{eq 2} ;

$$-y + x \sin x = -a \sin x - b \cos x$$

In eq (3) put the above eq value

$$y'' = (-y + x \sin x) + 2 \cos x - x \sin x$$

$$y'' = -y + 2 \cos x$$

$$y'' + y - 2 \cos x = 0$$

Answer

8)

$$y = a \sin(x+3)$$

Solution

$$y = a \sin x + 3a \sin$$

$$y' = -a \cos x - 3a \cos x$$

$$y' = a \cos(x+3)$$

$$a = \frac{y}{\sin(x+3)}$$

$$a = \frac{y'}{\cos(x+3)}$$

$$\frac{y}{\sin(x+3)} \neq \frac{y'}{\cos(x+3)}$$

$$y(\cos(x+3)) = y'(\sin(x+3))$$

$$y = \frac{y'(\sin(x+3))}{\cos(x+3)}$$

$$y = y' \tan(x+3)$$

$$y' = \frac{1}{\tan(x+3)} y$$

$$\underline{y' = y \cot(x+3)}$$

Answer

Initial / Boundary Problems

$y' = -\frac{x}{y}$, $y(3) = 4$ given that $x^2 + y^2 = r^2$ is the sol of DE

Solution

$$y' = -\frac{x}{y}$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

$$\int y dy = \int -x dx$$

$$\frac{y^2}{2} = -\frac{x^2}{2} + C$$

x by 2

$$y^2 = -x^2 + 2C$$

$$y^2 + x^2 = 2C$$

let $2C = r^2$

$$y^2 + x^2 = r^2 \quad \text{Answer}$$

$$3^2 + 4^2 = r^2$$

$$25 = r^2$$

$$x^2 + y^2 = 25$$

$$y^2 = \sqrt{25 - x^2}$$

$$y = \pm \sqrt{25 - x^2}$$

$$y = + \sqrt{25 - x^2} \quad \text{Answer}$$

34/b) $y'' - y' - 12y = 0$, $y(0) = -2$, $y'(0) = 6$ given that $y = ae^{4x} + be^{-3x}$ is the sol of given D.E

Solution

$$y'' - y' - 12y = 0$$

$$y = ae^{4x} + be^{-3x}$$

$$y' = 4ae^{4x} + (-3be^{-3x})$$

$$y = -2, \quad x = 0, \quad y' = 6$$

$$6 = 4ae^{4(0)} + (-3be^{-3(0)})$$

Substitute $x = 0$, $y = -2$

$$-2 = ae^{4(0)} + be^{-3(0)}$$

$$-2 = a + b \rightarrow 1$$

substitute $y' = 6, x = 0$

$$6 = 4ae^{4(0)} + -3be^{-3(0)}$$

$$6 = 4a - 3b \rightarrow 2$$

$$a = -2 - b$$

$$6 = 4(-2 - b) - 3b$$

$$6 = -8 - 4b - 3b$$

$$6 = -8 - 7b$$

$$14 = 7b$$

$$b = -2$$

$$a = -2 - (-2)$$

$$a = 0$$

$$y = 4(0)e^{4x} + (-2)e^{-3x}$$

$$\boxed{y = -2e^{-3x}} \quad \text{Answer}$$

) $y'' + y = 0, y(0) = 1, y'(\pi) = 1$ given that $y = a \sin x + b \cos x$

Solution

$$y = a \sin x + b \cos x$$

$$y' = a \cos x - b \sin x$$

substitute $x = 0, y = 1$

$$1 = a \sin(0) + b \cos(0)$$

$$1 = 0 + b$$

$$b = 1$$

substitute $x = \pi, y' = 1$

$$1 = a \cos \pi - b \sin \pi$$

$$1 = a(-1) - b(0)$$

$$1 = -a$$

$$a = -1$$

substitute $a = -1$, $b = 1$

$$y = -1 \sin x + 1 \cos x$$

$$y = \cos x - \sin x \quad \text{Answer}$$

SEPERABLE VARIABLE

$$x e^{x^2+y} dx = y dy$$

solution

$$x e^{x^2+y} dx = y dy$$

$$x e^{x^2} \cdot e^y dx = y dy$$

$$x e^{x^2} dx = y dy / e^y$$

$$\int x e^{x^2} dx = \int y dy e^{-y}$$

$$\int x dx e^{x^2} = \int y e^{-y} dy$$

$$\therefore u.v = u \int v + \int u' \int v$$

$$\text{let } z = x^2, dz = 2x dx$$

$$\int e^z x dx = y \int e^{-y} + \int y' \int e^{-y}$$

x is by

$$\frac{1}{2} \int e^z x dz = y - e^{-y} + -e^{-y}$$

$$\frac{1}{2} \int e^z dz = -e^{-y} (y + 1) + c$$

$$\frac{1}{2} e^z$$

$$\frac{1}{2} e^z = -e^{-y}(y+1) + e$$

$$\frac{1}{2} e^{x^2} = -2e^{-y}(y+1) + c \quad \text{Answer}$$

Q6 $(3x+8)(y^2+4)dx - 4y(x^2+5x+6)dy = 0$, $y(1) =$

Solution

$$(3x+8)(y^2+4)dx = 4y(x^2+5x+6)dy$$

$$\frac{(3x+8)}{(x^2+5x+6)} dx = \frac{4y}{y^2+4} dy$$

$$\int \frac{3x+8}{(x+3)(x+2)} dx = \int \frac{4y}{y^2+4} dy + c$$

using partial fractions

$$\frac{3x+8}{(x+3)(x+2)} = \frac{A}{x+3} + \frac{B}{x+2}$$

$$\frac{3x+8}{(x+3)(x+2)} = \frac{A(x+2) + B(x+3)}{(x+3)(x+2)}$$

$$3x+8 = A(x+2) + B(x+3)$$

$$x = -3$$

$$3(-3)+8 = A(-3+2) + B(-3+3)$$

$$1 = A \quad 0$$

$$x = -2$$

$$3(-2)+8 = A(-2+2) + B(-2+3)$$

$$2 = 0 + B(1)$$

$$B = 2$$

$$\frac{3x+8}{(x+3)(x+5)} = \frac{2}{x+2} + \frac{1}{x+3}$$

$$2 \int \frac{1}{x+2} + \int \frac{1}{x+3}$$

$$= 2 \ln(x+2) + \ln(x+3)$$

$$= \ln(x+2)^2 (x+3)$$

R.H.S

$$\int \frac{4y}{y^2+4} dy \rightarrow 2 \int \frac{2y}{y^2+4} \rightarrow 2 \ln(y^2+4) + C$$

$$\rightarrow \ln(y^2+4)^2$$

$$(x+2)^2(x+3) = C(y^2+4)^2$$

$$\text{put } y=2 \quad \& \quad x=1$$

$$(1+2)^2(1+3) = C(4+4)^2$$

$$9 \times 4 = C(64)$$

$$C = \frac{36}{64} \rightarrow \frac{9}{16}$$

$$(x+2)^2(x+3) = \frac{9}{16}(y^2+4)^2$$

Answer

$$y' = 2x^2 + y - x^2y + xy - 2x - 2$$

Solution

$$\frac{dy}{dx} = 2x^2 - 2x - 2 - x^2y + xy + y$$

$$\frac{dy}{dx} = 2x^2 - x^2y - xy - 2x + y - 2$$

$$y' = -x^2(-2+y) - x(y-2) + (y-2)$$

$$y' = (y-2)(-x^2 - x + 1)$$

$$\int \frac{dy}{y-2} = \int dx(-x^2 - x + 1)$$

$$\ln(y-2) = -\frac{x^3}{3} - \frac{x^2}{2} + x + C$$

Answer

$$\operatorname{cosec} y \, dx + \sec x \, dy = 0$$

Solution

$$\operatorname{cosec} y \, dx = -\sec x \, dy$$

$$\frac{\operatorname{cosec} y}{dy} = \frac{-\sec x \, dy}{dx}$$

$$\int \frac{1}{\operatorname{cosec} y} dy = \int \frac{1}{\sec x} dx + C$$

$$-\int \sin y \, dy = \int \cos x \, dx + C$$

$$\cos y = \sin x + C \quad \underline{\text{Ans}}$$

$$y \sqrt{1+x^2} dx + x \sqrt{1+y^2} dy = 0$$

$$\frac{y \sqrt{1+x^2}}{x \sqrt{1+y^2}} =$$

Solution

$$y \sqrt{1+x^2} dx + x \sqrt{1+y^2} dy = 0$$

$$y \sqrt{1+x^2} dx = -x \sqrt{1+y^2} dy$$

$$\int \frac{\sqrt{1+y^2}}{y} dy = - \int \frac{\sqrt{1+x^2}}{x} dx$$

Put $z = \sqrt{1+x^2}$, $z^2 = 1+x^2$, $2z dz = 2x dx$, $dx = z dz / x$

L.H.S

$$\int -\frac{z}{x} \cdot \frac{z dz}{x} = \int \frac{z^2}{x^2} dz \rightarrow \int \frac{z^2}{z^2-1} dz \rightarrow \int \frac{(z^2-1)+1}{z^2-1} dz$$

$$= \int 1 + \frac{1}{z^2-1} dz$$

$$= z + \frac{1}{2} \ln \left(\frac{z-1}{z+1} \right)$$

$$= \sqrt{1+x^2} + \frac{1}{2} \ln \left(\frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1} \right) \quad \text{R.H.S}$$

Similarly

$$= \sqrt{1+y^2} + \frac{1}{2} \ln \left(\frac{\sqrt{1+y^2}-1}{\sqrt{1+y^2}+1} \right) \quad \text{L.H.S}$$

$$\sqrt{1+y^2} + \frac{1}{2} \ln \left(\frac{\sqrt{1+y^2}-1}{\sqrt{1+y^2}+1} \right) + \sqrt{1+x^2} + \frac{1}{2} \ln \left(\frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1} \right) = C$$

Answer

Q10

$$\frac{dy}{dx} = \frac{y^3 + 2y}{x^2 + 3x}$$

Solution

$$\int \frac{1}{y(y^2+2)} dy = \int \frac{1}{x(x+3)} dx + \ln C$$

$$\frac{1}{y(y^2+2)} = \frac{A}{y} + \frac{By+C}{y^2+2}$$

$$\frac{1}{y(y^2+2)} = \frac{A(y^2+2) + y(By+C)}{(y)(y^2+2)}$$

$$1 = A(y^2+2) + y(By+C)$$

$$1 = Ay^2 + 2A + By^2 + Cy$$

$$y=0$$

$$1 = 0 + 2A + 0 + 0$$

$$A = \frac{1}{2}$$

$$y=1$$

$$1 = A + 2A + B + C$$

$$\frac{1}{y(y^2+2)} = \frac{\frac{1}{2}}{y} + \frac{-\frac{1}{2}y + 0}{y^2+2}$$

$$\int \frac{1}{y(y^2+2)} = \frac{1}{2} \int \frac{1}{y} dy - \frac{1}{2} \int \frac{2y}{y^2+2} dy$$

$$= \frac{1}{2} \left[\ln y - \frac{1}{2} \ln(y^2+2) \right]$$

$$= \frac{1}{2} \ln \frac{y}{\sqrt{y^2+2}}$$

$$\frac{1}{x(x+3)} = \frac{\frac{1}{3}}{x} + \frac{-\frac{1}{3}}{x+3}$$

$$\int \frac{1}{x(x+3)} dx = \frac{1}{3} \int \frac{1}{x} dx - \int \frac{1}{x+3} dx$$

$$= \frac{1}{3} \ln x - \ln(x+3)$$

$$= \frac{1}{3} \ln \frac{x}{x+3}$$

$$\frac{1}{2} \ln \frac{y}{\sqrt{y^2+2}} = \frac{1}{3} \ln \frac{x}{x+3} + \ln c \quad \text{Answer}$$

$$x^{2+y} 2x \cos y dx + x^2 (\sec y - \sin y) dy = 0$$

solution

$$-\int \frac{2x^3}{x^4} dx = \int \frac{\sec y - \sin y}{\cos y} dy$$

$$-2 \int \frac{1}{x} dx = \int \sec^2 y - \tan y dy$$

$$\frac{\sec y}{\cos y} = \frac{\sin y}{\cos y}$$

$$-2 \ln x + c = \tan y - \ln \sec y$$

$$Q_{12} (x+y)^2 (xy' + y) = xy(1+y')$$

Sol

$$\int \frac{xy' + y}{xy} dx = \int \frac{(1+y')}{(x+y)^2} dx$$

$$\ln xy = -1/(x+y) + c$$

$$\therefore \int [f(x)]^n f' dx = \frac{[f(x)]^{n+1}}{n+1}$$

$$(x+y)^{-2} \cdot (1+y') dx$$

$$= \frac{(x+y)^{-1}}{-2+1} \rightarrow \frac{(x+y)^{-1}}{-1}$$

Answer

$$\left(-\frac{1}{x+y} \right)$$

$$\frac{dy}{dx} = \cos(x+y)$$

$$\int \frac{1}{2} \frac{dz}{\cos^2 z} = \int 1 \cdot dx + c$$

Solution

Put $z = x+y$

$$\frac{dz}{dx} = \frac{dy}{dx} + \frac{dx}{dx} + 1$$

$$\frac{dz}{dx} = \frac{dy}{dx} + 1$$

$$\frac{dy}{dx} = \frac{dz}{dx} - 1$$

$$\frac{dz}{dx} - 1 = \cos z$$

$$\frac{dz}{dx} = \cos z + 1$$

$$\int \frac{dz}{\cos z + 1} = \int 1 \cdot dx$$

$$\left\{ \begin{aligned} \frac{1}{2} \int \sec^2(z/2) dz &= x + c \\ \frac{1}{2} \frac{\tan z/2}{1/2} &= x + c \end{aligned} \right.$$

$$\frac{1}{2} \tan\left(\frac{x+y}{2}\right) = x + c$$

$$\tan\left(\frac{x+y}{2}\right) = 2x + c$$

Answer

HOMOGENEOUS & REDUCIBLE TO HOMOGENEOUS

$$(x^2 - y^2)dx + 2xy dy = 0$$

Solution

$$(x^2 - y^2)dx = -2xy dy$$

$$\frac{dy}{dx} = \frac{x^2 - y^2}{2xy}$$

$$\text{let } y = vx, \quad v = \frac{y}{x}$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = \frac{x^2 - v^2 x^2}{2xvx}$$

$$v + x \frac{dv}{dx} = \frac{x^2(1 - v^2)}{x^2(2v)}$$

$$= \frac{1 - v^2}{2v} - v$$

$$= \frac{1 - v^2 - 2v^2}{2v}$$

$$x \frac{dv}{dx} = \frac{1 + 3v^2}{2v}$$

$$\int \frac{2v}{1 + 3v^2} dv = \int \frac{1}{x} dx$$

$$2 \int \frac{v}{1 + 3v^2} = \int \frac{1}{x} dx$$

% x by 3

$$\frac{1}{3} \int \frac{6v}{1 + 3v^2} = \ln x + \ln c$$

$$\ln(1 + 3v^2) = 3 \ln xc$$

$$\ln(1 + 3v^2) = -\ln(xc)^3$$

$$1 + 3v^2 = x^3 c$$

put the value of v

$$\ln \left(\frac{x^2 + 3y^2}{x^2} \right) = \ln \frac{c}{x^3}$$

$$\frac{x^2 + 3y^2}{x^2} = \frac{c}{x^3}$$

$$c = x \left[\frac{x^2 + 3y^2}{x^2} \right]$$

$$\boxed{c = x^2 + 3y^2} \quad \text{Answer}$$

Q2 $(x^2 + xy + y^2)dx - x^2dy = 0$

Solution

$$x(x^2 + xy + y^2) = x^2 dy$$

$$\frac{dy}{dx} = \frac{x^2 + xy + y^2}{x^2}$$

Put $y = vx$, $v = \frac{y}{x}$, $y' = v + \frac{x dv}{dx}$

$$v + \frac{x dv}{dx} = \frac{x^2 + xvx + v^2x^2}{x^2}$$

$$\frac{x dv}{dx} = \cancel{x^2} \left(\frac{1 + v + v^2}{\cancel{x^2}} \right) - v$$

$$\frac{x dv}{dx} = 1 + v + v^2 - v$$

$$\frac{x dv}{dx} = 1 + v^2$$

$$\int \frac{1}{1+v^2} = \int \frac{1}{x} dx$$

$$\tan^{-1} v = \ln Cx$$

$$\boxed{\tan^{-1} \left(\frac{y}{x} \right) = \ln Cx}$$

Answer

$$(1 + e^{x/y}) dx + e^{x/y} (1 - x/y) dy = 0$$

Solution

$$v = x/y, \quad x = vy, \quad \frac{dx}{dy} = v + y \frac{dv}{dy}$$

$$(1 + e^v)(v dy + y dv) + e^v(1 - v) dy = 0$$

$$\underbrace{((1 + e^v)v dy + e^v(1 - v)) dy}_{dy \text{ terms}} + \underbrace{(1 + e^v)y dv}_{dv \text{ terms}} = 0$$

$$v + y \frac{dv}{dy} = \frac{-e^{x/y} (1 - x/y)}{(1 + e^{x/y})}$$

$$v + y \frac{dv}{dy} = \frac{-e^v(1 - v)}{1 + e^v}$$

$$y \frac{dv}{dy} = \frac{-e^v(1 - v)}{1 + e^v} - v$$

$$y \frac{dv}{dy} = \frac{-e^v + v e^v - v(1 + e^v)}{1 + e^v}$$

$$y \frac{dv}{dy} = \frac{-e^v - v}{1 + e^v}$$

$$\int \frac{-1 + e^v}{v + e^v} dv = \int \frac{1}{y} dy$$

$$-\ln(v + e^v) = \ln y + \ln c$$

$$-\ln(v + e^v) = \ln(y c)$$

$$(v + e^v)^{-1} = y c$$

$$y_c = \frac{1}{v + e^v}$$

put the value

$$\boxed{\frac{1}{x/y + e^{x/y}} = y_c} \quad \text{Answer}$$

Q4

$$(2x + 3y + 4)dx - (4x + 6y + 5)dy = 0$$

Solution :

$$dx(2x + 3y + 4) = (4x + 6y + 5)dy$$

$$\frac{dy}{dx} = \frac{2x + 3y + 4}{4x + 6y + 5}$$

$$\text{here } \frac{a_1}{a_2} = \frac{b_1}{b_2}$$

$$\frac{dy}{dx} = \frac{(2x + 3y) + 4}{2(2x + 3y) + 5} \rightarrow \frac{1}{2}$$

$$\text{put } 2x + 3y = u$$

diff. w.r.t x

$$2 + 3\frac{dy}{dx} = \frac{du}{dx}$$

$$3\frac{dy}{dx} = \frac{du}{dx} - 2$$

$$\therefore \frac{dy}{dx} = \frac{1}{3} \left(\frac{du}{dx} - 2 \right)$$

$$\frac{1}{3} \left(\frac{du}{dx} - 2 \right) = \frac{(2x + 3y) + 4}{2u + 5}$$

$$\frac{du}{dx} - 2 = \frac{3u + 12}{2u + 5}$$

$$\frac{dy}{dx} - 2 = \frac{3u + 12}{2u + 5}$$

$$\frac{dy}{dx} = \frac{3u + 12}{2u + 5} + 2$$

$$\frac{dy}{dx} = \frac{3u + 12 + 4u + 10}{2u + 5}$$

$$\frac{dy}{dx} = \frac{7u + 22}{2u + 5}$$

$$\int \left(\frac{2u + 5}{7u + 22} \right) du = \int dx + c \longrightarrow 2$$

$$\frac{(7u + 22) \cdot 2u + 5 \left(\frac{2}{7} \right)}{\text{divisor } 7u \times \frac{2}{7}, 22 \times \frac{2}{7}}$$

$$\begin{array}{r} u + \frac{64}{7} \\ - \quad \quad \quad 5 - \frac{64}{7} \rightarrow \frac{35 - 64}{7} = -\frac{29}{7} \end{array} \left. \vphantom{\begin{array}{r} u + \frac{64}{7} \\ - \quad \quad \quad 5 - \frac{64}{7} \end{array}} \right\} R$$

$$\frac{2u + 5}{7u + 22} = \frac{2}{7} + \frac{-\frac{29}{7}}{7u + 22}$$

$$\text{eq 2} \rightarrow \int \left(\frac{2}{7} + \frac{-\frac{29}{7}}{7u + 22} \right) du = x + c$$

$$\frac{2}{7} \int du + \frac{-29}{7} \int \frac{1}{7u + 22} du = x + c$$

$$\frac{2}{7} u + -\frac{29}{7} \log |7u + 22| = x + c$$

$$\text{put } u = 2x + 3y$$

$$\frac{1}{7} (2x + 3y) + \frac{-29}{49} \log |7(2x + 3y) + 32| = x + c$$

$$\frac{4}{7}x + 3y - 29 \log |14x + 21y + 32| = x + c$$

$$\frac{4}{7}x + 3y - 29 \log |14x + 21y + 32| = 49x + \frac{49}{49}c$$

$$\frac{4}{7}x + 3y - 29 \log |14x + 21y + 32| = 49x + \frac{49}{49}c$$

$$-48x + 3y - 29 \log |14x + 21y + 32| = c' \quad \text{where } c' = 49c$$

$$3(y - 16x) + -29 \log (14x + 21y + 32) = c'$$

$$5 \quad y' = \frac{(x + y)}{x}, \quad y(1) = 1$$

Solution

$$y' = \frac{x}{x} + \frac{y}{x} \rightarrow y' = 1 + \frac{y}{x}$$

$$\frac{dy}{dx} = 1 + \frac{y}{x}$$

$$\text{let } y = vx, \quad v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = 1 + \frac{vx}{x}$$

$$x \frac{dv}{dx} = 1 + v - v$$

$$x \frac{dv}{dx} = 1$$

$$\int dv = \int dx \frac{1}{x}$$

$$v = \ln|x| + c$$

substitute v back

$$\frac{y}{x} = \ln|x| + c$$

$$y = x \ln|x| + Cx$$

$$1 = 1 \ln 1 + C(1)$$

$$1 = 0 + C$$

$$\boxed{C = 1} \rightarrow \boxed{y = x \ln x + x}$$

Answer

$$(4x + 6y + 3) dx = (6x + 9y + 2) dy$$

Solution

$$\frac{dy}{dx} = \frac{4x + 6y + 3}{6x + 9y + 2}$$

$$\frac{dy}{dx} = \frac{2(2x + 3y) + 3}{3(2x + 3y) + 2}$$

$$\text{Let } v = 2x + 3y$$

$$\therefore \frac{dy}{dx} = \frac{2v + 3}{3v + 2}$$

diff v w.r.t x

$$\frac{dv}{dx} = \frac{2}{3} \frac{dv}{dx}$$

$$\frac{dv}{dx} = 2 + 3 \frac{dy}{dx}$$

$$\frac{dv}{dx} - 2 = \frac{3dv}{dx} \cdot \frac{2v + 3}{3v + 2}$$

$$\frac{dv}{dx} = \frac{6v + 9}{3v + 2} + 2$$

$$\frac{dv}{dx} = \frac{6v + 9 + 2(3v + 2)}{3v + 2}$$

$$\frac{dv}{dx} = \frac{6v + 9 + 6v + 4}{3v + 2}$$

$$\frac{dv}{dx} = \frac{12v + 13}{3v + 2}$$

$$dx = \frac{3v + 2}{12v + 13} dv$$

$$\int \frac{3v + 2}{12v + 13} = \int dx$$

~~long division~~

$$\frac{1}{4} \int \frac{12v + 8 + 5 - 5}{12v + 13} dv = \int dx$$

$$\frac{1}{4} \int \frac{12v + 13}{12v + 13} dv - \frac{5}{4} \int 1 dv = \int dx$$

$$\frac{1}{4} \int 1 dv - \frac{5}{4} \int 1 dv = \int dx$$

$$-\int 1 dv = \int dx$$

$$v = -x + c$$

$$v + x = c$$

$$2x + 3y + x = c$$

$$3x + 3y = c$$

$$\boxed{3(x + y) = c} \text{ Ans}$$

Q1 $(x + 2y^3) \frac{dy}{dx} = y$

Solution

$$\frac{dy}{dx} = \frac{y}{x + 2y^3}$$

$$\frac{dx}{dy} = \frac{x + 2y^3}{y}$$

$$\frac{dx}{dy} - \frac{1}{y}x = 2y^2 \rightarrow 1$$

$$P(y) = 1, \text{ I.F.} = e^{-\int \frac{1}{y} dy} = e^{-\ln y} = y^{-1}$$

$\times$ by I.F

$$y^{-1} \left[\frac{dx}{dy} - \frac{1}{y}x \right] = 2y$$

$$\frac{d}{dy} [x \times \text{I.F.}] = 2y$$

$$xy^{-1} = 2 \int y dy + c$$

$$y^2 + c$$

$$\Rightarrow x = y(y^2 + c)$$

Answer

$$(1+x^2) \frac{dy}{dx} + 4xy = \frac{1}{(1+x^2)^2}$$

Solution

$$\frac{dy}{dx} + \frac{4x}{1+x^2} y = \frac{1}{(1+x^2)^3}$$

$$P(x) = \frac{4x}{1+x^2} \rightarrow I.F = e^{\int 2x/(1+x^2)} dx$$

$$= e^{2 \ln(1+x^2)}$$

$$= e^{\ln(1+x^2)^2}$$

$$= (1+x^2)^2$$

$$y \cdot (1+x^2)^2 = \int \frac{1}{(1+x^2)^2} dx + C$$

$$= \tan^{-1} x + C$$

$$y \cdot (1+x^2)^2 = \tan^{-1} x + C$$

Answer

Q3

$$\cos^3 x \frac{dy}{dx} + y \cos x = \sin x$$

Solution

∴ b.s by $\cos^3 x$

$$y' = \sec^2 x \cdot y = \frac{\sin x}{\cos^3 x} = \tan x \sec^2 x$$

$$P.N. = [\sec^2 x] \rightarrow I \cdot f = e^{\int \sec^2 x dx} \rightarrow e^{\tan x}$$

x eq 1 by I.f

$$e^{\tan x} \left[\frac{dy}{dx} + \sec^2 x \cdot y \right]$$

$$e^{\tan x} \tan x \sec^2 x$$

$$\frac{d}{dx} [y \times I \cdot f] = e^{\tan x} \tan x \sec^2 x$$

$$y \cdot e^{\tan x} = \int e^{\tan x} \tan x \sec^2 x \, dx + C$$

$$\therefore z = \tan x, \quad dz = \sec^2 x \, dx$$

$$y \cdot e^{\tan x} = \int z e^z dz + C$$

$$= z e^z - \int e^z dz + C$$

$$= z e^z - e^z + C$$

$$= e^z (z - 1) + C$$

$$y e^{\tan x} = e^{\tan x} (\tan x - 1) + C$$

$$\therefore \text{b.s by } \tan x \rightarrow y = (\tan x - 1) + C \cdot e^{-\tan x}$$

Answer

Case: HOMOGNEOUS Higher Order Differential Equation

$$(D^3 - 4D^2 + D + 6)y = 0$$

Solution

Auxiliary equation

$$m^3 - 4m^2 + m + 6 = 0$$

put $m = -1$

$$(-1)^3 - 4(-1)^2 + (-1) + 6 = 0$$

$$-1 - 4 - 1 + 6 = 0$$

$$0 = 0$$

	D^3	D^2	D	constant
-1	-1	-4	-1	+6
		-1	5	-6
	-1	-5	6	0

$$D^2 - 5D + 6 = 0$$

$$D^2 - 3D - 2D + 6 = 0$$

$$D(D-3) - 2(D-3) = 0$$

$$(D-2)(D-3) = 0$$

$$D-2=0, \quad D-3=0$$

$$D=2, \quad D=3$$

$\therefore D = -1, 2, 3$ real & distinct roots

$$y = c_1 e^{-x} + c_2 e^{2x} + c_3 e^{3x} \quad \text{Answer}$$

$$Q_2 \quad (4D^4 - 4D^3 - 3D^2 + 4D - 1)y = 0$$

Solution

Auxiliary equation

$$4m^4 - 4m^3 - 3m^2 + 4m - 1 = 0$$

put $m = 1$

$$4(-1)^4 - 4(1)^3 - 3(1)^2 + 4(1) - 1 = 0$$

$$4 - 4 - 3 + 4 - 1 = 0$$

$$0 = 0$$

1	4	-4	-3	4	-1
		4	0	-3	1
4	0	-3	1	0	

$$4m^3 - 3m + 1 = 0$$

put $m = -1$

$$4(-1)^3 - 3(-1) + 1 = 0$$

$$-4 + 3 + 1 = 0$$

$$0 = 0$$

-1	4	0	-3	1
		-4	4	-1
4	-4	1	0	

$$4m^2 - 4m + 1 = 0$$

$$(2r - 1)^2 = 0$$

$$r = \frac{1}{2}, \quad (r-1)(r+1)(2r-1)^2 = 0$$

$$\text{roots} = 1, -1, \frac{1}{2}, \frac{1}{2}$$

$$y = C_1 e^x + C_2 e^{-x} + (C_3 + C_4 x) e^{x/2} \quad \text{Answer}$$

$$(D^4 + 2D^3 - 2D^2 - 6D + 5)y = 0$$

Solution

Auxiliary equation

$$m^4 + 2m^3 - 2m^2 - 6m + 5 = 0$$

$$\text{put } m = 1$$

$$(1)^4 + 2(1)^3 - 2(1)^2 - 6(1) + 5 = 0$$

$$1 + 2 - 2 - 6 + 5 = 0$$

$$0 = 0$$

1	1	2	-2	-6	5
		1	3	1	-5
	1	3	1	-5	0

$$(r-1)(r^3 + 3r^2 + r - 5) = 0$$

$$\text{Put } r = 1$$

$$(1)^3 + 3(1)^2 + 1 - 5 = 0$$

$$1 + 3 + 1 - 5 = 0$$

$$0 = 0$$

1	1	3	1	-5
		1	4	-5
	1	4	5	0

$$r^2 + 4r + 5 = 0$$

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$r = \frac{-4 \pm \sqrt{4^2 - 4(1)(5)}}{2(1)}$$

$$r = \frac{-4 \pm \sqrt{16 - 20}}{2}$$

$$r = \frac{-4 \pm \sqrt{-4}}{2}$$

$$r = \frac{-4 \pm 2i}{2}$$

$$r = -2 \pm i$$

$$\text{roots} = 1, 1, -2+i, -2-i$$

$$y = (c_1 + c_2 x)e^x + (c_3 \sin x + c_4 \cos x)e^{-2x} \quad \text{Answer}$$

Case: By Exponential

$$4 \quad (D^2 - 3D + 2)y = e^{3x}$$

Solution :

Auxiliary equation

$$D^2 - 3D + 2 = 0, \quad m^2 - 3m + 2 = 0$$

$$D^2 - 1 - 2 + 2 = 0$$

$$(D - 1)(D - 2) = 0$$

$$D - 1 = 0, \quad D - 2 = 0$$

$$D=1, D=2 \leftarrow \text{roots}$$

$$y_c = C_1 e^x + C_2 e^{2x}$$

$$P.I = \frac{1}{f(D)} Q(x)$$

$$P.I = \frac{1}{D^2 - 3D + 2} \times e^{3x}$$

$$P.I = \frac{1}{3^2 - 3(3) + 2} \times e^{3x} \neq 0$$

$$P.I = \frac{1}{9 - 9 + 2} \times e^{3x}$$

$$P.I = \frac{1}{2} e^{3x}$$

$$y = C.F + P.I$$

$$y = C_1 e^x + C_2 e^{2x} + \frac{1}{2} e^{3x} \quad \text{Answer}$$

$$Q5 \quad (D^2 + 3D - 4)y = 15 e^{2x}$$

Solution:

Auxiliary equation

$$m^2 + 3m - 4 = 0$$

$$m^2 + 4m - m - 4 = 0$$

$$m(m+4) - 1(m+4) = 0$$

$$(m+4)(m-1) = 0$$

$$m = -4, m = 1$$

$$y_c = c_1 e^x + c_2 e^{-4x}$$

$$y_p = \frac{1}{D^2 + 3D - 4} \times (15e^{2x})$$

$$y_p = \frac{1}{(2)^2 + 3(2) - 4} \times 15e^{2x}$$

$$y_p = \frac{1}{4 + 6 - 4} \times 15e^{2x}$$

$$= \frac{5}{2} e^{2x}$$

$$y = y_c + y_p$$

$$y = c_1 e^x + c_2 e^{-4x} + \frac{5}{2} e^{2x}$$

Q6 $(D^3 - 2D^2 - 5D + 6)y = e^{3x}$

Solution

Auxiliary equation

$$m^3 - 2m^2 - 5m + 6 = 0$$

Put $m = 1$

$$(1)^3 - 2(1)^2 - 5(1) + 6 = 0$$

$$1 - 2 - 5 + 6$$

$$0 = 0$$

1	1	-2	-5	+ 6
		1	-1	-6
	1	-1	-6	0

$$m^2 - m - 6$$

$$m^2 - 3 + 2 - 6$$

$$(m-3)(m+2)$$

$$m = 3$$

$$m = -2$$

$$\text{roots} = 1, 3, -2$$

$$y_c = c_1 e^x + c_2 e^{-2x} + c_3 e^{3x}$$

$$y_p = \frac{1}{(D^3 - 2D^2 - 5D + 2)} e^{3x}$$

$$= \frac{x}{3D^2 - 4D - 5} e^{3x}$$

$$= \frac{x}{3(3)^2 - 4(3) - 5}$$

$$= \frac{x}{27 - 12 - 5} e^{3x}$$

$$= \frac{x e^{3x}}{10}$$

$$y = y_c + y_p$$

$$y = c_1 e^x + c_2 e^{-2x} + c_3 e^{3x} + \frac{x e^{3x}}{10}$$

Answer

Case: By Trigonometric ratios

Q7 $(D^2 - 5D + 6)y = \sin 2x$

Solution:

Auxiliary equation

$$m^2 - 5m + 6 = 0$$

$$m^2 - 3 - 2m + 6$$

$$m(m-3) - 2(m-3)$$

$$(m-2)(m-3)$$

$$m = 2, m = 3$$

$$y_c = C_1 e^{2x} + C_2 e^{3x}$$

$$y_p = \frac{1}{D^2 - 5D + 6} \sin 2x$$

$\therefore$ replace $D^2 = -a^2$

$$y_p = \frac{1}{-2^2 - 5D + 6} \sin 2x$$

$$y_p = \frac{1}{-4 - 5D + 6} \sin 2x$$

$$y_p = \frac{1}{2 - 5D} \times \frac{2 + 5D}{2 + 5D} \sin 2x$$

$$y_p = \frac{2 + 5D}{4 - 25D^2} \sin 2x$$

$$y_p = \frac{2 + 5D}{4 - 25(-2^2)}$$

$$= \frac{2 + 5D}{4 + 100} \sin 2x$$

$$= \frac{1}{104} (2 + 5D) \sin 2x$$

$$= \frac{1}{104} \left(2 \sin 2x + 5 \frac{d}{dx} (\sin 2x) \right)$$

$$= \frac{1}{104} (2 \sin 2x + 5(2 \cos 2x))$$

$$= \frac{1}{104} (2 \sin 2x + 10 \cos 2x)$$

$$= \frac{1}{52} (\sin 2x + 5 \cos 2x)$$

$$y = c_1 e^{2x} + c_2 e^{3x} + \frac{1}{52} (\sin 2x + 5 \cos 2x) \quad \text{Answer}$$

$$(D^2 - 2D + 4)y = e^x \cos x$$

Solution:

Auxiliary eq

$$m^2 - 2m + 4 = 0$$

$$m = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$m = \frac{-2 \pm \sqrt{2^2 - 4(1)(4)}}{2(1)}$$

$$m = \frac{2 \pm \sqrt{4 - 16}}{2}$$

$$m = \frac{2 \pm \sqrt{-12}}{2}$$

$$m = \frac{2 \pm 2i\sqrt{3}}{2}$$

$$m = 1 \pm i\sqrt{3}$$

$$y_c = e^x (c_1 \sin \sqrt{3}x + c_2 \cos \sqrt{3}x) \quad D = D+1 \text{ (1)}$$

$$y_p = \frac{1}{D^2 - 2D + 4} (e^x \cos x) \rightarrow e^x \cdot \frac{1}{D^2 - 2D + 4} \cos x$$

$$y_p = \frac{1}{(D+1)^2 - 2(D+1) + 4} e^x \cos x \quad e^x \frac{1}{(D+1)^2 - 2(D+1) + 4} \cos x$$

$$y_p = e^x \cos x \cdot \frac{1}{D^2 + 3}$$

$$e^x \frac{1}{D^2 + 2D + 1 - 2D - 2 + 4} \cos x$$

$$y_p = e^x \cdot \frac{1}{-1^2 + 3} \cos x$$

$$= \frac{e^x \cos x}{2}$$

$$y = y_c + y_p$$

$$y = e^x (c_1 \sin \sqrt{3}x + c_2 \cos \sqrt{3}x) + (e^x \cos x) \frac{1}{2}$$

Answer

$$(D^2 - 2D - 3)y = 2e^x - 10\sin x$$

Solution:

$$m^2 - 2m - 3 = 0$$

$$m^2 - 3 + 1m - 3 = 0$$

$$m(m-3) + 1(m-3)$$

$$(m+1)(m-3)$$

$$m = -1, m = 3$$

$$y_c = C_1 e^{-x} + C_2 e^{-3x}$$

$$y_p = \frac{1}{D^2 - 2D - 3} [2e^x - 10\sin x]$$

$$y_p = \frac{1}{D^2 - 2D - 3} 2e^x - \frac{1}{D^2 - 2D - 3} 10\sin x$$

$$= 2 \frac{1}{1 - 2 - 3} e^x - 10 \frac{1}{(-1)^2 - 2D - 3} \sin x$$

$$= -\frac{e^x}{2} + \frac{10}{2(D+2)} \times \frac{(D-2)}{(D-2)} \sin x$$

$$= -\frac{e^x}{2} + \frac{5(D-2)}{D^2 - 4} \sin x$$

$$= -\frac{e^x}{2} + \frac{5(D-2)}{(-1)^2 - 4} \sin x$$

$$y_p = -\frac{e^x}{2} + \frac{5(D\sin x - 2\sin x)}{-5}$$

$$= -\frac{e^x}{2} - (\cos x - 2\sin x)$$

$$y_c + y_p = y$$

$$y = C_1 e^{-x} + C_2 e^{3x} - \frac{e^x}{2} - (\cos x - 2\sin x)$$

Answer

Case : by Algebraic / constant

Q10

$$y'' - 2y' + 2y = x$$

Solution :

Auxiliary equation

$$m^2 - 2m + 2 = 0$$

$$m = \frac{\pm b \sqrt{b^2 - 4ac}}{2a}$$

$$m = \frac{\pm(-2) \sqrt{(-2)^2 - 4(1)(2)}}{2(1)}$$

$$m = \frac{2 \pm \sqrt{4-8}}{2}$$

$$m = \frac{2 \pm \sqrt{-4}}{2}$$

$$m = \frac{2 \pm 2i}{2}$$

$$m = 1 \pm i$$

$$y_c = e^x (c_1 \cos x + c_2 \sin x)$$

$$y_p = \frac{1}{D^2 - 2D + 2} x \Rightarrow$$

$$= \frac{1}{2 \left(1 + \frac{D^2}{2} - D \right)} x$$

$$2 \left(\frac{D^2}{2} - \frac{2D}{2} + \frac{2}{2} \right)$$

$$= \frac{1}{2 \left[1 - \left(D - \frac{D^2}{2} \right) \right]} x$$

$$= \frac{1}{2} \underbrace{\left[1 - \left(D - \frac{D^2}{2} \right) \right]}_1^{-1} x$$

$$\therefore \frac{1}{(1-x)} \Rightarrow (1-x)$$

$$= 1 - x + x^2 - x^3 + \dots$$

$$= \frac{1}{2} \left[1 - \left(D - \frac{D^2}{2} \right) + \left(D - \frac{D^2}{2} \right)^2 - \dots \right] x$$

$$= \frac{1}{2} \left[1 - D + \frac{D^2}{2} + D^2 - \frac{2D^2}{2} + \frac{D^4}{4} - \dots \right] x$$

$$= \frac{1}{2} [1 - D] x$$

$$= \frac{1}{2} (x - Dx)$$

$$y_p = \frac{1}{2} (x - 1)$$

$$y = y_c + y_p$$

$$y = e^x (c_1 \cos x + c_2 \sin x) + \frac{1}{2} (x - 1) \quad \text{Answer}$$

Q11 $(D^3 + D)y = 2x^2 + 4\sin x$

Solution

Auxiliary eq

$$m^3 + m = 0$$

$$m(m^2 + 1) = 0 \rightarrow m^2 = -1 \rightarrow m = \pm \sqrt{-1} \rightarrow \pm i$$

$$m = 0, m^2 + 1 = 0$$

$$m = 0, m = +i, m = -i$$

$$y_c = C_1 + C_2 \cos x + C_3 \sin x$$

$$y_p = y_{p1} + y_{p2} \rightarrow \frac{1}{D^3 + D} (2x^2 + 4\sin x)$$

$$y_p = \frac{1}{D(D^2 + 1)} (2x^2 + 4\sin x)$$

$$= \frac{1}{D^2 + 1} \left(\frac{2x^3}{3} - 4\cos x \right)$$

$$= \frac{1}{D^2 + 1} \left(\frac{2x^3}{3} \right) - \frac{1}{D^2 + 1} (4\cos x)$$

$$= [1 + D^2]^{-1} \left(\frac{2x^3}{3} \right) - \frac{x}{2D} (4\cos x)$$

$$= (1 - D^2) \left(\frac{2x^3}{3} \right) - 2\sin x$$

$$= \frac{2}{3}x^3 - 4x - 2\sin x$$

$$y = C_1 + C_2 \cos x + C_3 \sin x + \frac{2}{3}x^3 - 4x - 2\sin x \quad \text{Answer}$$

$$y'' - 4y = (2-3)x \quad ; \quad y(0) = 0 \quad , \quad y'(0) = 5$$

Solution:

$$m^2 - 4 = 0$$

$$m^2 = 4$$

$$m = \pm 2$$

$$y_c = c_1 e^{2x} + c_2 e^{-2x}$$

$$y_p = Ax + B$$

$$y'_p = A$$

$$y''_p = 0$$

$$y'' - 4y = 2 - 3x$$

$$0 - 4(Ax + B) = 2 - 3x$$

$$-4Ax - 4B = 2 - 3x$$

$$-4A = -3$$

$$A = \frac{3}{4}$$

$$-4B = 2$$

$$B = -\frac{1}{2}$$

$$y_p = \frac{3}{4}x - \frac{1}{2}$$

$$y(x) = y_c + y_p$$

$$= c_1 e^{2x} + c_2 e^{-2x} + \frac{3}{4}x - \frac{1}{2}$$

$$\text{Given } y(0) = 0, \quad y'(0) = 5$$

Taking derivative of $y(x)$ /general solution

$$y'(x) = 2c_1 e^{2x} - 2c_2 e^{-2x} + 2$$

$$y(0) = 0$$

$$y(0) = c_1 e^0 + c_2 e^0 + 2(0) - \frac{1}{2}$$

$$0 = c_1 + c_2 - \frac{1}{2}$$

$$c_1 + c_2 = \frac{1}{2} \rightarrow 5$$

$$y'(0) = 5$$

$$y'(0) = 2c_1 e^0 - 2c_2 e^0 + 2$$

$$5 = 2c_1 - 2c_2 + 2$$

$$3 = 2c_1 - 2c_2 \rightarrow 3,$$

$$c_1 - c_2 = \frac{3}{2} \rightarrow 4,$$

add eq 1 into 2, $\times$ eq 1 by 2

$$c_1 + c_2 = \frac{1}{2}$$

$$2c_1 + 2c_2 = 1 \rightarrow \text{eq 1}$$

$$(2c_1 + 2c_2) + (2c_1 - 2c_2) = 1 + 3$$

$$4c_1 = 4$$

$$\boxed{c_1 = 1}$$

put c_1 into eq 5

$$c_1 + c_2 = \frac{1}{2}$$

$$-1 + \frac{1}{2} = c_2$$

$$\boxed{c_2 = -\frac{1}{2}}$$

$$\boxed{y(x) = e^{2x} - \frac{1}{2}e^{-2x} + 2x - \frac{1}{2}} \quad \text{A}$$